Quantum Resonance Unified Model (QRUM)

Foundational Hypothesis, Scientific Research Programme,
Authorship and Priority Documentation

Author: P. Gabor (Gábor Palásti)

Affiliation: Independent Researcher, Hungary

Date: 13 October 2026 | **Document type:** Foundational Scientific Statement and Research Provenance Record

Version: 1.3

1. Origin and Foundational Scientific Hypothesis

The Quantum Resonance Unified Model (QRUM), also developed under the designation Einstein–Quantum Resonance Spacetime Extension (EQRSE), is an independent theoretical and phenomenological research programme initiated by P. Gabor.

The first public disclosures of the programme were deposited on Zenodo on **11 November 2025**, including the documents titled *Einstein – P. Gabor Quantum Resonance Derivation: Unified Physical Model of Space–Time–Energy* and *Einstein – Quantum Resonance Unified Model (Summary Version)*. Subsequent early deposits followed on 16 November 2025 (*Einstein–Quantum Resonance Spacetime Extension*, DOI: **10.5281/zenodo.17620325**; concept DOI for all versions of this record: **10.5281/zenodo.17618387**) and 20 November 2025.

These November 2025 deposits constitute the earliest time-stamped public records of the QRUM research programme and establish the documentary starting point of the work.

The original motivation of the programme can be expressed through two conceptual analogies that guided the early formulation:

"What if it is not the lamp that produces the light, but the light that illuminates the lamp?"

Consider a tree. Its mass, structure and gravitational influence can be examined through established physical laws. A larger tree presents different physical quantities and gravitational effects. Another question can be asked: what conditions make the existence of the tree possible? Water, soil, air, sunlight and suitable environmental interactions provide the conditions under which a tree may develop.

The QRUM hypothesis proposes an analogous question in fundamental physics: could deeper, potentially shared frequency-dependent structures and resonance conditions contribute to the emergence, organization or observable properties of physical systems?

Within this conceptual interpretation, resonance may be both a consequence of physical organization and a possible contributing factor in shaping that organization.

These analogies express the original motivation of the research programme. They are not mathematical derivations or experimental evidence.

2. Relationship to Einstein's General Relativity

QRUM is intended to investigate whether additional frequency-dependent phenomenological descriptions could complement established gravitational physics.

Einstein's general relativity describes the relationship between spacetime geometry and energy-momentum. QRUM does not reject this established framework.

The underlying research question is whether measurable frequency-dependent residuals, correlations or organizational structures might reveal additional physical relationships not fully characterized by the baseline models used in particular analyses.

The currently published QRUM Phase 1 residual framework introduces a phenomenological spectral response function:

$$f_\phi(f) = 1 + \alpha / [1 + ((f - f_0)/(f_0/Q))^2]$$

Here, α represents a fitted response amplitude, f_0 a characteristic frequency and Q a quality factor.

This expression is a spectral test function, not a derived fundamental law of nature. It does not presently constitute a generally covariant modification of Einstein's field equations.

The deeper hypothesis concerning resonance and physical organization remains a theoretical research direction requiring further mathematical development and independent experimental testing.

3. Research Development and Experimental Directions

Following the initial November 2025 disclosures, the QRUM research programme has developed through several related investigations:

- **Gravitational-wave residual analysis:** Phenomenological searches for weak frequency-dependent spectral features in publicly available gravitational-wave observations, including baseline subtraction, instrumental-noise controls and reproducibility testing. The current published results are null or inconclusive and do not establish a new physical resonance.
- **Astrophysical frequency populations:** Statistical comparisons of characteristic frequencies across compact-object classes, including white dwarfs, neutron stars, pulsars and other astrophysical systems. These studies examine reproducible population-level structures without presupposing a common physical mechanism.
- **Environmental dependence:** Investigation of pulsar spin frequencies in relation to model-estimated Galactic dark-matter density. A quality-filtered sample of 4,184 pulsars yielded a statistically detectable but very weak population-level association, with Spearman ρ approximately 0.054. This result does not demonstrate direct physical coupling or validate QRUM.
- **Optical atomic-clock testing:** A proposed independent metrological approach to constrain hypothetical weak frequency-dependent residuals through high-precision frequency comparisons. Sensitivity scales approaching 10^{-18}–10^{-19} describe experimental capabilities and requirements, not a confirmed QRUM signal prediction.
- **Multi-modal correlation-based sensing:** An exploratory approach to identifying weak perturbative signatures through phase, frequency, coherence and temporal correlations across independent measurement channels.

These investigations represent connected but scientifically distinguishable components of the broader QRUM programme. **None individually establishes the foundational hypothesis as a confirmed physical theory.**

4. Authorship, Scientific Provenance and Priority

P. Gabor (Gábor Palásti) identifies himself as the originator and author of the QRUM research programme and its original conceptual formulation.

The purpose of this statement is to provide a clear, dated record of that authorship claim, the foundational scientific hypothesis and its relationship to previously disclosed research.

The earliest public time-stamped records of the programme are the Zenodo deposits of **11 November 2025**. A subsequent early deposit of 16 November 2025 carries the DOI **10.5281/zenodo.17620325** (concept DOI for all versions of that record: **10.5281/zenodo.17618387**). Further development is documented through dated manuscripts, public research repositories, source-code records, research identifiers and submissions to the Hungarian Intellectual Property Office (SZTNH).

The earlier documentation includes the following references:

- SZTNH scientific-work submission: SZTNH_EBEJ_20251110-1684
- SZTNH patent-related submission: SZTNH_EBEJ_20251110-1645
- Hungarian patent application reference: P2500452

These identifiers are included for documentary traceability. Their inclusion does not independently establish registration validity, patent grant, enforceable patent claims or legal ownership of a scientific principle. The applicable legal status must be determined from the corresponding official records.

The author requests appropriate scholarly attribution when subsequent research materially relies upon, reproduces, extends or discusses specific original QRUM formulations, methods or published results.

Such attribution should accurately distinguish between the original hypothesis, methodological contributions, independent experimental verification and subsequent theoretical developments.

5. Intellectual Property and Legal Clarification

This document records an assertion of authorship and scientific provenance. It is not a declaration of exclusive ownership over physical laws, natural phenomena, general scientific ideas or independently developed discoveries.

Copyright protection, where applicable, concerns original protected expression, including qualifying manuscript text, figures and software, rather than scientific hypotheses or mathematical principles as such.

Patent rights, where applicable, depend on the legally established scope, status, jurisdiction and enforceability of particular patent claims.

Publication dates, DOI records, archived versions and source-code histories may provide evidence of prior public disclosure. They do not automatically establish exclusive scientific priority for every later related result or guarantee recognition as the discoverer of a subsequently verified phenomenon.

This statement does not replace any earlier manuscript, intellectual-property filing, licence or legally operative document, and does not amend their terms.

Independent researchers remain free to investigate related scientific questions. Where established scholarly standards require citation of relevant prior work, appropriate attribution is expected.

6. Public Research Records and Persistent Identifiers

Research author: P. Gabor (Gábor Palásti)

ORCID: https://orcid.org/0009-0009-9842-7814
Research website: https://gabor8107.github.io/
GitHub repository: https://github.com/gabor8107/QRUM
Open Science Framework (OSF): https://osf.io/4x2ht/ DOI: 10.17605/OSF.IO/4X2HT

Earliest public deposits (November 2025):

- 11 November 2025 — Einstein – P. Gabor Quantum Resonance Derivation
- 11 November 2025 — Einstein – Quantum Resonance Unified Model (Summary Version)
- 16 November 2025 — Einstein–Quantum Resonance Spacetime Extension — DOI: 10.5281/zenodo.17620325 (concept DOI: 10.5281/zenodo.17618387)

Selected later Zenodo publications:

- Einstein–QRUM / Residual Framework: https://doi.org/10.5281/zenodo.21796869
- QRUM Astrophysical Frequency Survey: https://doi.org/10.5281/zenodo.22050657
- QRUM Environmental Dependence: https://doi.org/10.5281/zenodo.21694558
- Optical Atomic-Clock Research: https://doi.org/10.5281/zenodo.22133035
- QRUM Multi-Modal Research: https://doi.org/10.5281/zenodo.21656162

Google Scholar: Bibliographic indexing and author-profile association, where available, serve as supplementary discovery and citation mechanisms. Google Scholar indexing is not a legal certification of authorship, priority or scientific validity.

7. Final Statement

The Quantum Resonance Unified Model represents an independently initiated research programme motivated by the possibility that resonance and deeper frequency-dependent structures may contribute to the organization and observable properties of physical systems.

Its objective is to investigate whether this foundational idea can be developed into mathematically specified, experimentally distinguishable and independently falsifiable physical predictions.

The programme seeks to complement established physics through testable investigation, not to replace established results through unsupported assertions.

This document consolidates the author's stated conceptual contribution, the earliest public disclosures of 11 November 2025, subsequent published research directions and documentary provenance as of 13 October 2026.

P. Gabor (Gábor Palásti)

Independent Researcher, Hungary

13 October 2026

End of statement.